Practical #3

Practical Title:
To apply GitHub-based collaboration techniques including branching and merging

Theory:

1. What is GitHub Collaboration?

GitHub collaboration is the process where multiple developers work together on the same project by sharing code, tracking changes, reviewing contributions, and managing versions using repositories, branches, and pull requests

2. What is Branching in Git?

Branching is the process of creating a **separate line of development** in a repository. It allows developers to work on new features or bug fixes without affecting the main code.

3. What is Merging?

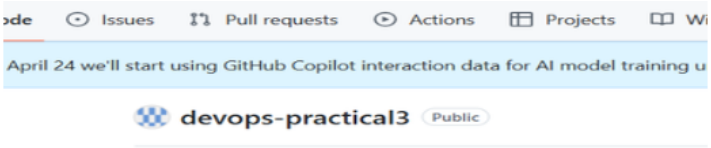
Merging is the process of **combining changes from one branch into another**, usually from a feature branch into the main branch.

4. What is Pull Request (PR)?

A Pull Request (PR) is a feature in GitHub where a developer **requests to merge changes** from one branch into another. It allows team members to review, discuss, and approve changes before merging.

Procedure:

Step 1: Create a GitHub Repository



Step 2: Clone Repository

```
cd devops-practical3
Cloning into 'devops-practical3'...
warning: You appear to have cloned an empty repository.
```

Step 3: Create a New Branch

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical3 (feature1)
$ git branch feature1
git checkout feature1
fatal: a branch named 'feature1' already exists
Already on 'feature1'
```

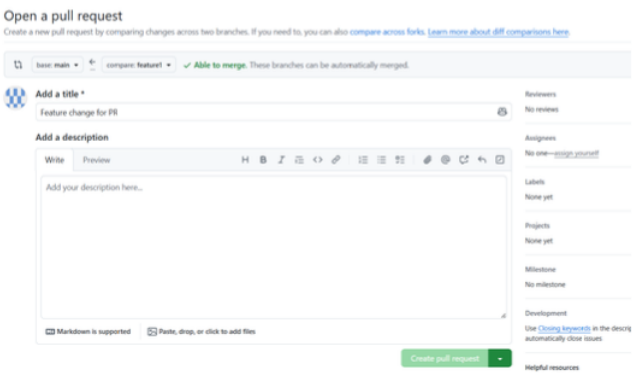
Step 4: Make Changes and Commit

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical3 (feature1)
$ echo "Collaboration work" > file.txt
git add .
git commit -m "Added feature work"
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
[feature1 98c6fad] Added feature work
 1 file changed, 1 insertion(+), 1 deletion(-)
```

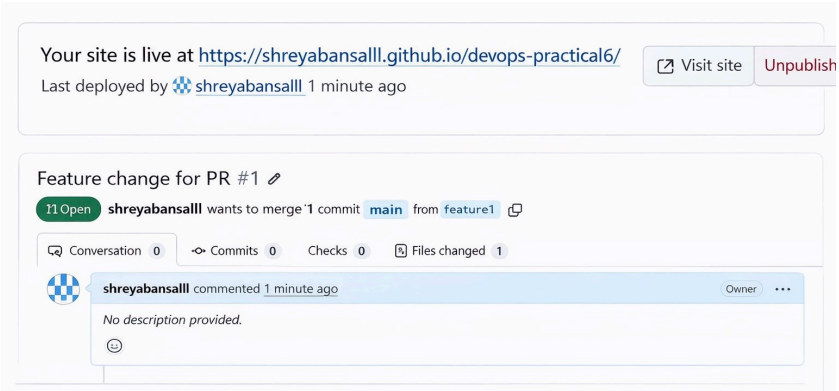
Step 5: Push Branch to GitHub

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical3 (feature1)
$ git push origin feature1
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 454 bytes | 113.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```

Step 6: Create Pull Request on GitHub



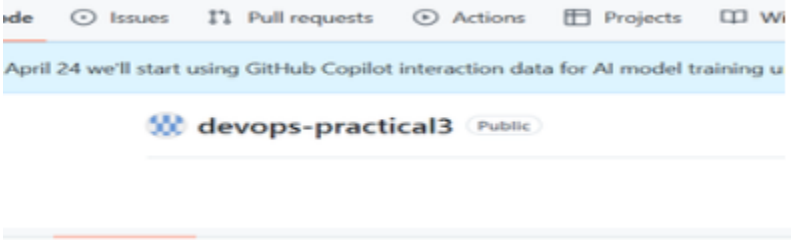
Step 7: Merge Pull Request

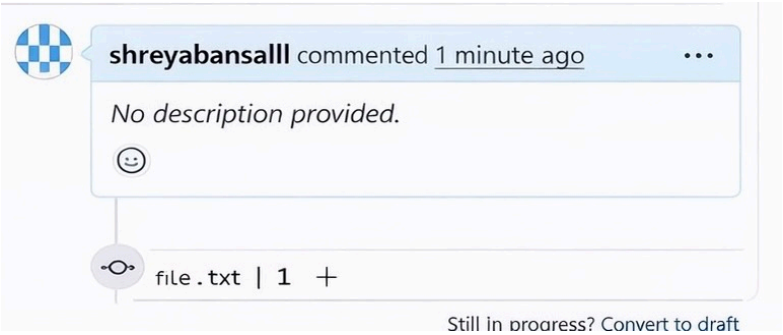


Step 8: Pull Updated Code

Activity / Task:

Task 1:
Create a repository and collaborate with at least one teammate.





Repository Link:
https://github.com/shreyabansalll/devops-practical3

Task 2:
Write the steps to resolve merge conflicts.

1. Pull the latest changes from the repository
2. Git will show conflicting files
3. Open the file and identify conflict markers:

   <<<<<<< HEAD
   your changes
   =======
   incoming changes
   >>>>>>> branch-name

4. Edit the file and keep the correct code
5. Remove conflict markers
6. Add the resolved file:

   git add file.txt

7. Commit the changes:

   git commit -m "Resolved merge conflict"

8. Push the changes:

   git push

Advantages of GitHub Collaboration:

1. Enables teamwork and code sharing
2. Easy tracking of changes
3. Code review through pull requests
4. Better project management
5. Supports distributed development

Challenges:

1. Merge conflicts
2. Requires coordination among team members
3. Learning Git commands can be difficult